



Hall Ticket No. -

Code: S123533

Stanley College of Engineering & Technology for Women (A)

B.E (ECE & EEE) III Semester (Backlog) Examinations-May-June-2023

Probability Theory and Stochastic Processes

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. What are the conditions for a function to be a random variable? [2M CO1 BTL1]
b. Specify the significance of standard deviation? [2M CO3 BTL2]
c. The density function of a random variable X is given as
 $f_X(x) = Ae^{-\left(\frac{x}{3}\right)}u(x)$, where $A > 0$. Find the constant A if $f_X(x)$ is valid [2M CO1 BTL3]
d. Differentiate time average and statistical average. [2M CO4 BTL2]
e. Write units for Autocorrelation function and power density spectrum. [2M CO5 BTL1]
f. Discuss the properties of Gaussian random variables. [2M CO2 BTL2]

Part-B

4X12=48M

2. a) A man wins a gambling game if he gets two heads in five flips of a biased coin. The probability of getting a head with the coin is 0.6. What is the probability of winning if he wins by getting at least four heads in five flips? Should he play this new game? [6M CO1 BTL5]
b) State Baye's Theorem and Total Probability. Identify the properties of probability Density Function. [6M CO1 BTL2]
3. a) Discuss the properties of joint density function of random variables X and Y. [8M CO2 BTL2]
b) Obtain mean, median, mode for the following data:

Value(x)	0	1	2	3	4	5
Frequency(f)	9	10	12	17	28	34

[4M CO3 BTL3]

4. Random variables X_1 and X_2 have joint characteristic function

$$\Phi_{X_1, X_2}(\omega_1, \omega_2) = [(1 - j2\omega_1)(1 - j2\omega_2)]^{-N/2} \quad \text{where } N > 0 \text{ is an integer.}$$

What is the correlation coefficient?

[12M CO2 BTL4]

5. a) Fit a straight line to the following data:

X	0	5	10	15	20	25
Y	12	15	17	22	24	30

[6M CO3 BTL4]

- b) Fit a parabola to the following data:

X	0	1	2	3	4
Y	1	1.8	1.3	2.5	6.3

[6M CO3 BTL4]

6. Given the random processes $X(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$ where ω_0 is a constant, A and B are uncorrelated zero-mean random variables having different density functions but the same variance σ^2 , show that $X(t)$ is a wide-sense stationary but not strictly stationary.

[12M CO4 BTL5]

7. a) If $X(t)$ is a stationary process find the power spectrum of $Y(t) = A + BX(t)$ in terms of power spectrum of $X(t)$ if A and B are real constants.

[8M CO5 BTL5]

- b) A stationary random process $X(t)$ is applied to the input of a system for which $h(t) = 3u(t)t^2 \exp(-8t)$. If $E[X(t)] = 2$, what is the mean value of the system's response $Y(t)$?

[4M CO5 BTL6]



Hall Ticket No. -

Code: S123538

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Backlog) Examinations-May-June-2023

Analog Electronics

Time: 03Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Draw PN Diode V-I Characteristics. [2M CO1 BTL2]
b. Explain early effect. [2M CO2 BTL2]
c. List the advantages of negative feedback Circuit. [2M CO3 BTL4]
d. Categorize the Power Amplifiers. [2M CO4 BTL4]
e. Define Barkhausen criterion. [2M CO5 BTL1]
f. What are the important characteristics of an Ideal operational amplifier? [2M CO4 BTL1]

Part-B

4X12=48M

2. a) Compare Intrinsic and extrinsic Semiconductors and PN junction diode and Zener diode. [6M CO1 BTL2]
b) Draw and explain about Centre Tapped Full wave Rectifier. [6M CO1 BTL2]
3. a) Discuss about Thermal Stability, S , S' , S'' of a transistor. [6M CO2 BTL6]
b) Explain the construction and operation of JFET. [6M CO2 BTL3]
4. a) Discuss about Voltage Shunt feedback amplifiers. [6M CO3 BTL6]
b) Compare positive and negative feedback concepts. [6M CO3 BTL4]
5. a) With a neat Diagram explain Hartley Oscillator. [6M CO4 BTL4]
b) With a neat Diagram explain Crystal Oscillator. [6M CO4 BTL4]
6. a) Explain the operation of Integrator Circuit using OP-AMP. [6M CO5 BTL2]
b) Discuss about Sample and Hold Circuit. [6M CO5 BTL3]
7. a) Explain the effect of negative feedback on input and output impedances. [6M CO3 BTL2]
b) Discuss about photo Diode with diagram. [6M CO1 BTL2]



Hall Ticket No. -

Code: S1235310

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Backlog) Examinations-May-June--2023

Electrical Circuit Analysis

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

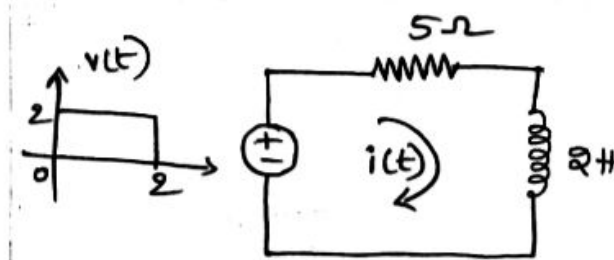
6X2=12M

1. a. Define RMS value. [2M CO1 BTL1]
- b. State Thevenin's and Norton's Theorem. [2M CO2 BTL2]
- c. What is meant by transient and steady state responses? [2M CO3 BTL2]
- d. List any three properties of Laplace transform. [2M CO4 BTL2]
- e. What is meant by driving point impedance? [2M CO5 BTL2]
- f. Define Quality Factor & Bandwidth. [2M CO1 BTL1]

Part-B

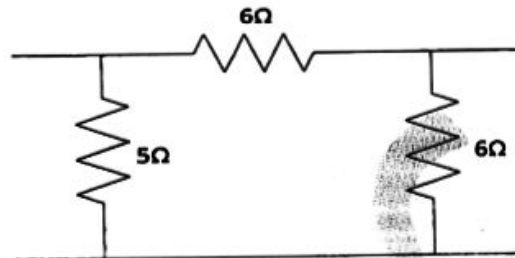
4X12=48M

2. Explain the following:
 - a) Series Resonance and Resonant Frequency. [6M CO1 BTL2]
 - b) Short notes on series aiding and opposing of magnetically coupled coils. [6M CO1 BTL2]
3. a) State and explain Superposition Theorem in detail. [6M CO2 BTL2]
- b) State and explain Thevenin's theorem in detail. [6M CO2 BTL2]
4. a) Derive the expression for current in a series RL circuit excited by DC supply by closing the switch at $t = 0^+$. [6M CO3 BTL3]
- b) Derive the expression for voltage in a series RC circuit excited by DC supply by closing the switch at $t = 0^+$. [6M CO3 BTL3]
5. a) State and prove Convolution Theorem using Laplace Transform. [6M CO4 BTL3]
- b) Find $i(t)$ for the circuit shown below using Laplace Transform. [6M CO4 BTL4]



6. a) Derive the relationship between hybrid and transmission parameters. [6M CO5 BTL4]

b) For the circuit below, determine impedance parameters [6M CO5 BTL3]



7. a) State and Explain Maximum power transfer theorem in detail. [6M CO2 BTL3]

b) Prove that the Line Voltage is equal to $\sqrt{3}$ times the Phase Voltage in a star connected 3- ϕ system.. [6M CO1 BTL3]



Hall ticket No. -

Code: S1235314

Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Backlog) Examinations-May-June-2023

Electro Magnetic Fields

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. What does pointing vector signify? [2M CO1 BTL1]
b. State Coulomb's law. [2M CO2 BTL2]
c. Define Magnetic flux and magnetic flux density. [2M CO3 BTL1]
d. Describe displacement current. [2M CO4 BTL3]
e. Define Skin Effect. [2M CO5 BTL4]
f. State Biot-Savart's Law. [2M CO3 BTL4]

Part-B

4X12=48M

2. a) Explain vector multiplication and triple products. [6M CO1 BTL4]
b) Calculate the cylindrical coordinates from the given spherical coordinates $r=5$, $\theta=60^\circ$, $\phi=30^\circ$ [6M CO1 BTL3]
3. a) State and Prove Gauss's law as applied to electro static fields. [6M CO2 BTL2]
b) A charge of 6nc is distributed uniformly along a line of length 6m . Find the field intensity at a radial distance of 2m from the centre of the line. The medium is air for which $\epsilon_r=1$ [6M CO2 BTL3]
4. a) Write a short notes on magnetic boundary conditions. [6M CO3 BTL2]
b) Explain force on a moving charge and force between differential current elements. [6M CO3 BTL4]
5. a) State and Explain point form of Ampere's circuit law. [6M CO3 BTL4]
b) Calculate the magnetic flux density due to a coil of 100 A and area 50cm^2 .
i) On the axis of a coil at a distance of 10m from centre and
ii) At a point 10m in a direction at right angles to the axis. [6M CO3 BTL5]
6. a) Discuss and derive in detail about the modified form of Maxwell's equation. [6M CO4 BTL3]
b) State and Explain Faraday's law of Electromagnetic Induction. [6M CO4 BTL4]
7. a) Explain Maxwell equations in phasor form. [6M CO5 BTL5]
b) Explain briefly about displacement current and significance of pointing vector. [6M CO5 BTL4]

Stanley College of Engineering & Technology for Women (A)**B.E (EEE) III Semester (Backlog) Examinations – May-June-2023****Signals and Systems Analysis****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Categorize signals and systems as seen in everyday life. [2M CO1 BTL4]
 b. Is the system described by the equation $y(t) = 2x(t)$ time invariant or not? Why? [2M CO2 BTL5]
 c. State any two properties of Fourier series. [2M CO3 BTL2]
 d. Evaluate the DTFT of $x(n) = u(n) - u(n-3)$. [2M CO4 BTL5]
 e. Solve for the Laplace transform of $e^{-3t}u(-t)$. [2M CO5 BTL3]
 f. State the properties of time scaling property & differentiation in Z-domain property. [2M CO5 BTL2]

Part-B**4X12=48M**

2. a) Distinguish between the (i) Continuous time signal and discrete time signal
 (ii) Periodic and Aperiodic Signals. [6M CO1 BTL2]
 b) Illustrate the reconstruction of the signal from its samples. [6M CO1 BTL4]
3. a) Explain about stability and causality of an LTI system. [6M CO2 BTL3]
 b) Solve for the convolution sum of two sequence using graphical method

$$x[n] = \{4, 1, 2, 5\} \quad h[n] = \{1, 2, -1\}$$

$\uparrow$
 0

$\uparrow$
 0

[6M CO4 BTL3]
4. a) Derive the expression for trigonometric Fourier series coefficients. [6M CO3 BTL3]
 b) Obtain the Fourier transform of the (i) Unit step function (ii) Unit impulse function. [6M CO3 BTL4]
5. a) State and prove Parseval's theorem. [6M CO4 BTL4]
 b) Solve for the 4-Point DFT of the given sequence $x(n) = \{1, -2, 3, 4\}$ [6M CO4 BTL3]

$\uparrow$
 0
6. a) Find the Z-transform of $x(n) = \cos(n\omega)u(n)$. [6M CO5 BTL4]
 b) Determine the Laplace transform of the signals (i) $x_1(t) = \cos(\omega_0 t)$ (ii) $x_2(t) = te^{-t}u(t)$. [6M CO5 BTL4]
7. Solve the given differential equation using Laplace transform

$$d^2y(t)/dt^2 + 3dy(t)/dt + 2y(t) = dx(t)/dt$$

Initial conditions $y(0^-) = 2$, $dy(0^-)/dt = 1$ and $x(t) = e^{-t}u(t)$; $x(0)$. Find the total Response of the system. [12M CO5 BTL3]
